

COMMUNICATION BETWEEN PSYCHOPY AND UNICORN: LSL TRIGGERS

Psychopy setup

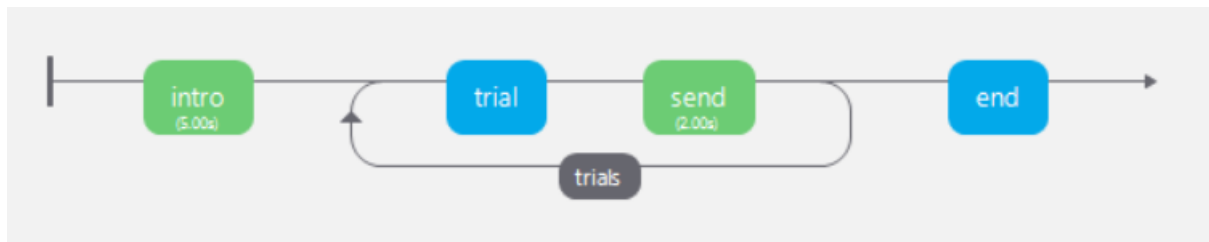
<https://mbraintrain.com/how-to-send-lsl-triggers-with-psychopy-and-mobile-eeg/>

1. Install the pylsl in order to make this LSL code work.

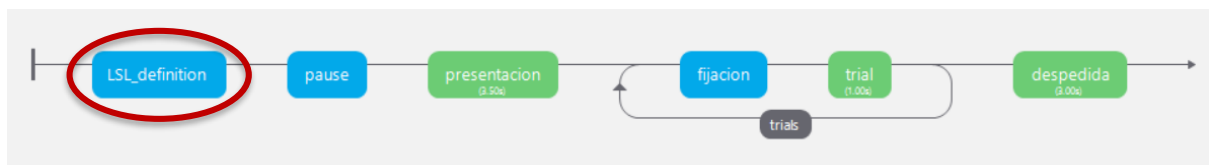
```
pip install pylsl
```

```
conda install -c conda-forge liblsl
```

2. Prepare a computerized cognitive experiment with PsychoPy (p.e. Motor Imagery):



3. Insert a new routine where we can define the LSL stream:



4. In this routine, add block of code.



5. Double click on “code” and define the LSL stream by indicating the next python code on the sections “Before experiment” and “Begin experiment”:

code Properties

Name: code Code Type: Auto->JS ☐ disabled

Before Experiment * Begin Experiment * Begin Routine Each Frame End Routine

```
1 from pylsl import StreamInfo, StreamOutlet
```

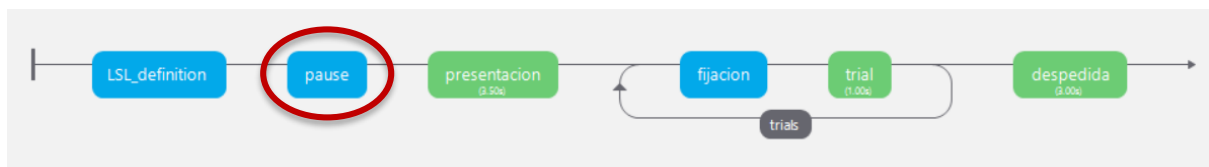
code Properties

Name: code Code Type: Auto->JS ☐ disabled

Before Experiment * Begin Experiment * Begin Routine Each Frame End Routine End Experiment

```
1 info = StreamInfo(name='PsychoPy_LSL', type='Markers',
2 channel_count=1, nominal_srate=0, channel_format='string',
3 source_id='psy_marker')
4
5 outlet = StreamOutlet(info) # Broadcast the stream.
```

6. Introduce a pause between LSL definition and the rest of the computarized task:



7. Add another block of code whenever the stimulus is sent and write the following code under the “Begin Routine” tab (SEND).

Outlet.push_sample(['event_name'])



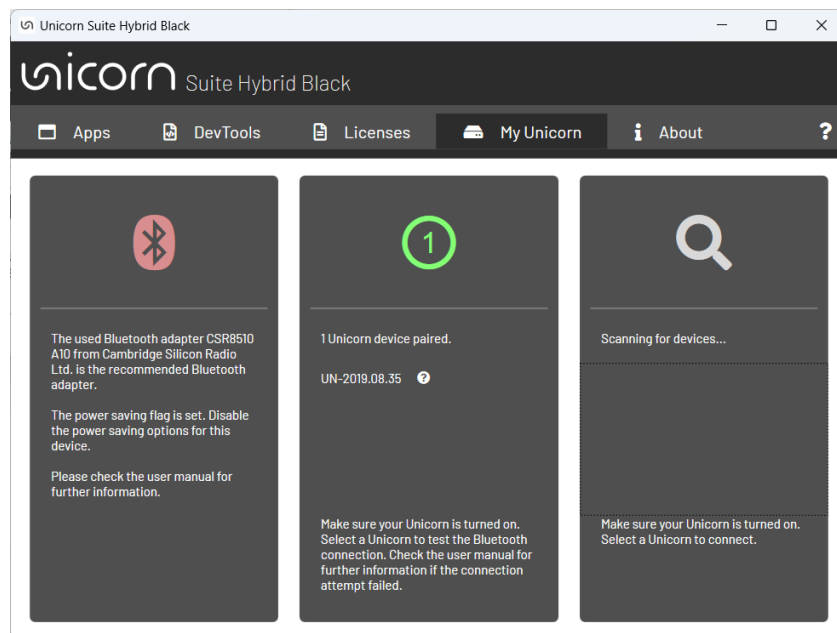
Before Experiment Begin Experiment Begin Routine * Each Frame End Routine End Experiment

```
1 outlet.push_sample(Direccion_flecha[9])
2 print(Direccion_flecha[9])
```

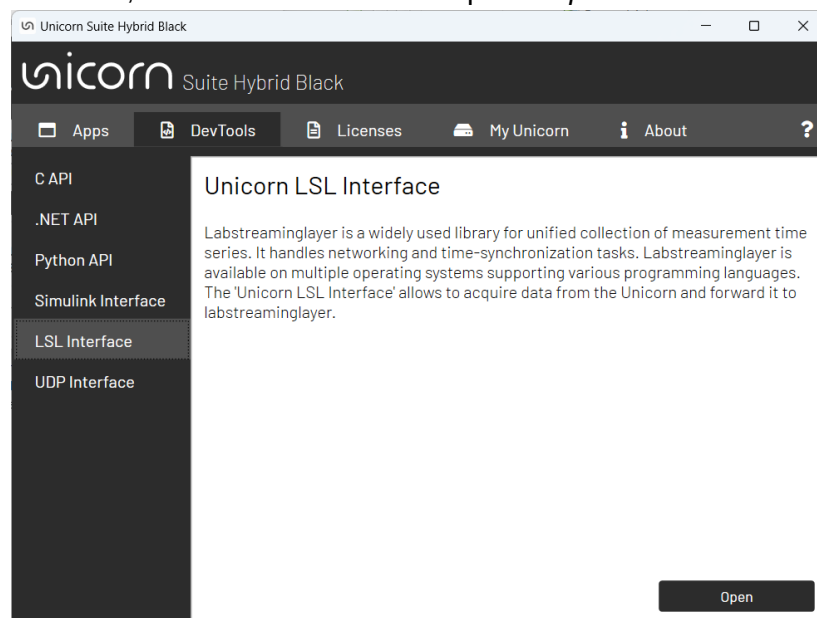
8. Make sure there is a pause after defining the LSL and before the experiment begins so that LabRecorder can detect the stream correctly. When LabRecorder has begun, then continue.

Unicorn LSL Setup

1. Open Unicorn Suite software and pair the Unicorn device to the laptop via Bluetooth (use adapter).



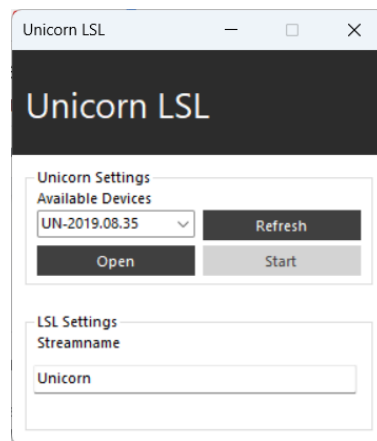
2. Under *DevTools*, select *LSL Interface* and press *Open*.



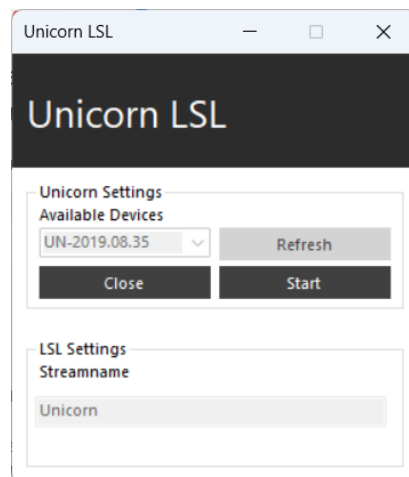
3. Execute *UnicornLSL* file.

Documents > gtec > Unicorn Suite > Hybrid Black > Unicorn LSL	
Name	Date modified
 liblsl64.dll	10/9/2020 15:45
 Unicorn.dll	30/7/2021 8:08
 UnicornDotNet.dll	30/7/2021 8:08
 UnicornLSL	30/7/2021 8:25

4. Select the Unicorn device and set a stream name for the EEG signals. Then, press *Open*.

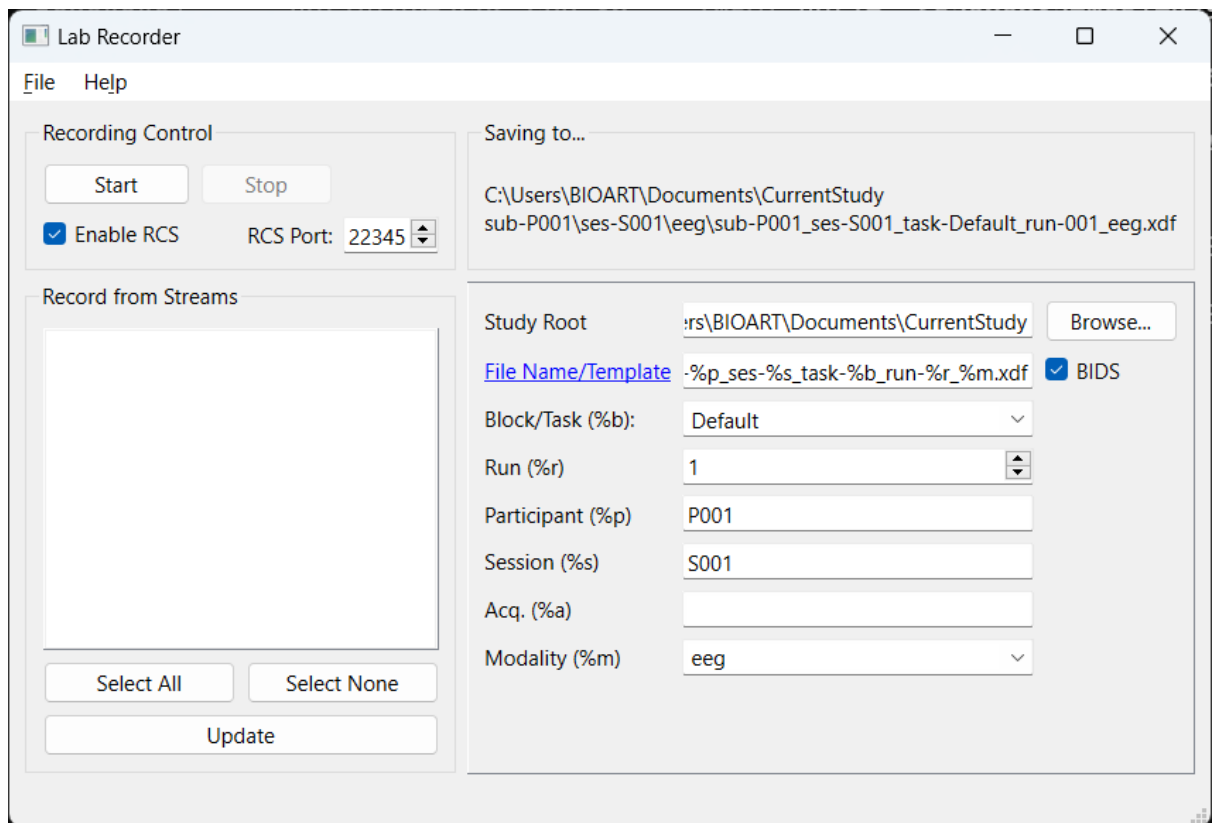


5. Start streaming.



Lab Recorder

1. Install software <https://github.com/labstreaminglayer/App-LabRecorder/releases>
2. A window like this will show up



3. Select streams to record (both events and eeg streams MUST be opened) and press *Start*.
4. When pressing *Stop*, an .xdf file will be saved on the *StudyRoot* path.

Matlab Processing

1. Use this module to open the dataset with Matlab [<https://github.com/xdf-modules/xdf-Matlab/tree/0cdf054391fff7f0ea3416ee632ad1bd73d6623b>]
 - I have upload the required script ('load_xdf.m') on Atenea.
2. `[streams,fileheader] = load_xdf('name_eeg_data.xdf')`